The Coh Bridge Theorem

From Verifier-Admissible Traces to Projection-Induced Geometry

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Abstract

We formulate the Coh Bridge Theorem, a formal connection between three layers of the Coh framework. The V1 layer constructs morphisms from verifier-admissible traces satisfying a resource inequality. The V2 layer adds hidden realizations, observable projection, and a fiberwise envelope defect. The V3 layer defines directed distance by taking the infimum of envelope defect over admissible traces. The bridge theorem states that verifier closure and additive bookkeeping yield a category; projection compatibility, fiber decomposition, and hidden cost subadditivity yield an oplax defect law; and the infimum construction turns that oplax law into the directed triangle inequality of a Lawvere-style quasi-metric. Thus the geometry of Coh is not imposed externally: it is induced by admissibility, projection, and envelope cost.

1 Purpose and Positioning

The Coh framework is organized as a layered construction:

$$\boxed{V1: \text{ verifier admissibility} \quad \Longrightarrow \quad V2: \text{ projection-certified category} \quad \Longrightarrow \quad V3: \text{ induced geometry.}}$$

The purpose of this paper is not to introduce a new fourth layer. It is to make the implication chain precise. The bridge theorem explains why the V3 metric enrichment is forced by the V2 oplax cost law, and why the V2 oplax cost law is the projection-aware lift of the V1 category of admissible traces.

The guiding principle is:

$$\boxed{\text{morphisms are earned by certification, not assumed as primitive arrows.}}$$

A transition becomes an arrow only after verifier acceptance and satisfaction of a resource law. Projection then assigns semantic cost by taking a worst-case bound over hidden realizations. Finally, minimizing this certified semantic cost over all admissible traces produces a directed geometry.

2 V1: Operational Admissibility

2.1 State space, valuation, and traces

Definition 2.1 (Operational state space)**.** *An operational state space is a set X. Elements of X are called observable states.*

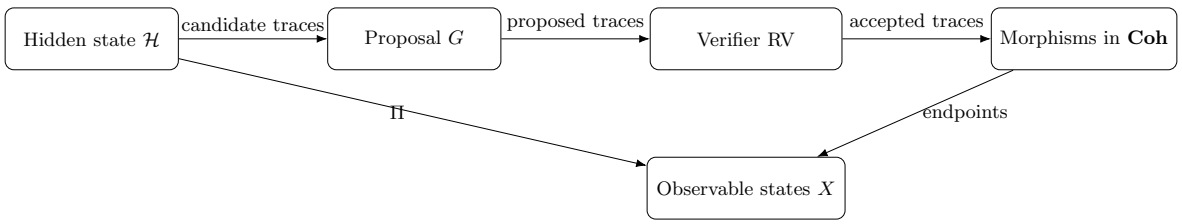


Figure 1: Operational wedge: hidden proposals enter a verifier; only accepted traces become morphisms in the Coh layer.

Definition 2.2 (Valuation)**.** *A valuation is a function*

$$V : X \to \mathbb{Q}_{\geq 0}$$

or, in analytic variants,

$$V : X \to \mathbb{R}_{\geq 0}.$$

The value $V(x)$ measures risk, violation, unresolved obligation, or remaining potential at state x.

Definition 2.3 (Trace)**.** *A finite trace $\tau : x \rightsquigarrow y$ is a finite composable sequence*

$$x = x_0 \xrightarrow{\sigma_1} x_1 \xrightarrow{\sigma_2} \cdots \xrightarrow{\sigma_n} x_n = y.$$

The empty trace at x is denoted $\epsilon_x : x \rightsquigarrow x$.

Definition 2.4 (Spend and defect)**.** *Each primitive step σ carries two nonnegative bookkeeping values:*

$$s(\sigma), d(\sigma) \in \mathbb{Q}_{\geq 0}.$$

For a trace $\tau = (\sigma_1, \ldots, \sigma_n)$ define

$$\mathrm{Spend}(\tau) := \sum_{i=1}^{n} s(\sigma_i), \qquad \mathrm{Defect}(\tau) := \sum_{i=1}^{n} d(\sigma_i).$$

The empty trace has zero spend and zero defect.

Lemma 2.5 (Additive bookkeeping)**.** *For composable traces $\tau_1 : x \rightsquigarrow y$ and $\tau_2 : y \rightsquigarrow z$,*

$$\mathrm{Spend}(\tau_2 * \tau_1) = \mathrm{Spend}(\tau_1) + \mathrm{Spend}(\tau_2),$$

and

$$\mathrm{Defect}(\tau_2 * \tau_1) = \mathrm{Defect}(\tau_1) + \mathrm{Defect}(\tau_2).$$

Proof. Trace concatenation is list append. The spend and defect functions are finite sums over the list of steps, so the identities follow from additivity of finite sums over appended lists. □

2.2 Verifier admissibility

Definition 2.6 (Deterministic verifier)**.** *A deterministic verifier is a function*

$$\mathsf{RV} : \mathrm{Trace}(X) \to \{\mathsf{ACCEPT}, \mathsf{REJECT}\}.$$

A trace τ is verifier-accepted when $\mathsf{RV}(\tau) = \mathsf{ACCEPT}$.

Axiom 2.7 (Verifier identity)**.** *For every state $x \in X$,*

$$\mathsf{RV}(\epsilon_x) = \mathsf{ACCEPT}.$$

Axiom 2.8 (Verifier closure)**.** *If $\tau_1 : x \rightsquigarrow y$ and $\tau_2 : y \rightsquigarrow z$ satisfy*

$$\mathsf{RV}(\tau_1) = \mathsf{ACCEPT}, \qquad \mathsf{RV}(\tau_2) = \mathsf{ACCEPT},$$

then

$$\mathsf{RV}(\tau_2 * \tau_1) = \mathsf{ACCEPT}.$$

Remark 2.9. *Verifier closure is a real structural assumption, not a formality. A context-sensitive verifier may accept two traces individually while rejecting their composite. If that happens, one obtains a graph or partial category of certified traces, not the V1 Coh category defined below.*

Definition 2.10 (Coh admissibility)**.** *A trace $\tau : x \rightsquigarrow y$ is Coh-admissible if*

$$\mathsf{RV}(\tau) = \mathsf{ACCEPT}$$

and

$$\boxed{V(y) + \mathrm{Spend}(\tau) \leq V(x) + \mathrm{Defect}(\tau).}$$

Remark 2.11 (Spend–defect separation)**.** *Spend and defect are not interchangeable. Spend is paid transition cost. Defect is allowed slack, tolerance, or uncertainty. Collapsing them into one scalar removes the distinction between controlled cost and tolerated deviation.*

2.3 The V1 category theorem

Definition 2.12 (V1 hom-set)**.** *For states $x, y \in X$, define*

$$\mathrm{Hom}_{\mathbf{Coh}_{V1}}(x, y) := \{\tau : x \rightsquigarrow y : \tau \text{ is Coh-admissible}\}.$$

Composition is trace concatenation:

$$\tau_2 \circ \tau_1 := \tau_2 * \tau_1.$$

The identity at x is $\mathrm{id}_x := \epsilon_x$.

Lemma 2.13 (Identity admissibility)**.** *For every state x, the empty trace ϵ_x is Coh-admissible.*

Proof. Verifier acceptance holds by the verifier identity axiom. Also

$$\mathrm{Spend}(\epsilon_x) = 0, \qquad \mathrm{Defect}(\epsilon_x) = 0.$$

Thus the resource law becomes

$$V(x) + 0 \leq V(x) + 0,$$

which holds by equality. □

Lemma 2.14 (Closure of admissibility)**.** *If $\tau_1 : x \rightsquigarrow y$ and $\tau_2 : y \rightsquigarrow z$ are Coh-admissible, then $\tau_2 * \tau_1 : x \rightsquigarrow z$ is Coh-admissible.*

Proof. Verifier acceptance of the composite follows from verifier closure. It remains to prove the resource law. Since τ_1 and τ_2 are admissible,

$$V(y) + \mathrm{Spend}(\tau_1) \leq V(x) + \mathrm{Defect}(\tau_1),$$

and

$$V(z) + \mathrm{Spend}(\tau_2) \leq V(y) + \mathrm{Defect}(\tau_2).$$

Adding gives

$$V(y) + V(z) + \mathrm{Spend}(\tau_1) + \mathrm{Spend}(\tau_2) \leq V(x) + V(y) + \mathrm{Defect}(\tau_1) + \mathrm{Defect}(\tau_2).$$

Assuming finite intermediate valuation, cancel $V(y)$ to obtain

$$V(z) + \mathrm{Spend}(\tau_1) + \mathrm{Spend}(\tau_2) \leq V(x) + \mathrm{Defect}(\tau_1) + \mathrm{Defect}(\tau_2).$$

By additive bookkeeping,

$$V(z) + \mathrm{Spend}(\tau_2 * \tau_1) \leq V(x) + \mathrm{Defect}(\tau_2 * \tau_1).$$

Thus the composite is admissible. □

Theorem 2.15 (V1 category theorem)**.** *Assume verifier identity, verifier closure under concatenation, finite cancellable valuations, and additive trace bookkeeping. Then Coh-admissible traces form a category* $\mathbf{Coh}_{V1}$.

Proof. Identities are admissible by the identity admissibility lemma. Composition is closed by the closure of admissibility lemma. Associativity follows from associativity of list concatenation:

$$(\tau_3 * \tau_2) * \tau_1 = \tau_3 * (\tau_2 * \tau_1).$$

The empty trace gives the unit laws:

$$\epsilon_y * \tau = \tau, \qquad \tau * \epsilon_x = \tau.$$

Therefore the category axioms hold. □

$$\begin{tikzcd} x \arrow[r, "\tau_1"] \arrow[rr, bend left, "\tau_2 \circ \tau_1 = \tau_2 * \tau_1"] & y \arrow[r, "\tau_2"] & z \end{tikzcd}$$

Figure 2: V1 composition: certified composition is trace concatenation.

3 V2: Projection and Envelope Defect

3.1 Hidden and observable layers

Definition 3.1 (Hidden substrate)**.** *A hidden substrate consists of a class of hidden realizations $\mathcal{G}$ and a partially defined concatenation operation*

$$\cdot : \mathcal{G} \times \mathcal{G} \rightharpoonup \mathcal{G}.$$

When $\xi_2 \cdot \xi_1$ is defined, ξ_1 is executed first and ξ_2 second.

Definition 3.2 (Observable trace substrate)**.** *An observable substrate consists of a class of observable traces $\mathcal{V}$, a partially defined composition operation*

$$\odot : \mathcal{V} \times \mathcal{V} \rightharpoonup \mathcal{V},$$

and an identity trace $e \in \mathcal{V}$.

Definition 3.3 (Projection-cost layer)**.** *A projection-cost layer consists of a projection map*

$$\Pi : \mathcal{G} \to \mathcal{V}$$

and a hidden cost functional

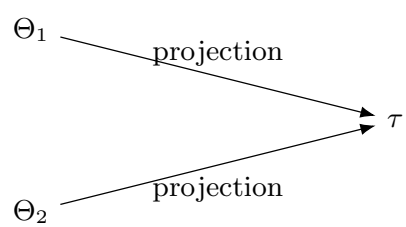
$$W : \mathcal{G} \to \mathbb{R}_{\geq 0}.$$

Definition 3.4 (Fiber)**.** *For $R \in \mathcal{V}$, define*

$$\mathcal{F}_R := \{\xi \in \mathcal{G} : \Pi(\xi) = R\}.$$

Definition 3.5 (Envelope defect)**.** *For an admissible observable trace R with nonempty bounded fiber, define*

$$\boxed{\delta(R) := \sup_{\xi \in \mathcal{F}_R} W(\xi).}$$



multiple hidden realizations can collapse to one observable trace

Figure 3: Projection collapse: multiple hidden realizations may project to one observable trace. The envelope defect takes the worst-case hidden cost over the fiber.

3.2 Axioms for the projection bridge

Axiom 3.6 (Projection compatibility)**.** *If $\xi_2 \cdot \xi_1$ is defined, then*

$$\Pi(\xi_2 \cdot \xi_1) = \Pi(\xi_2) \odot \Pi(\xi_1).$$

Axiom 3.7 (Fiber decomposition)**.** *If $\xi \in \mathcal{G}$ satisfies*

$$\Pi(\xi) = R_2 \odot R_1,$$

then there exist $\xi_1, \xi_2 \in \mathcal{G}$ such that

$$\xi = \xi_2 \cdot \xi_1, \qquad \Pi(\xi_1) = R_1, \qquad \Pi(\xi_2) = R_2.$$

Remark 3.8 (Segmentable witness reading)**.** *Fiber decomposition should not be interpreted as a universal property of arbitrary hidden dynamics. It is a theorem for segmentable witness types: a composite hidden witness must carry enough boundary data to split into component witnesses. In continuous or memory-bearing systems, this axiom must either be proved from an interval-indexed witness structure or replaced by a weak glue-defect law.*

Axiom 3.9 (Hidden cost subadditivity)**.** *If $\xi_2 \cdot \xi_1$ is defined, then*

$$W(\xi_2 \cdot \xi_1) \leq W(\xi_1) + W(\xi_2).$$

If equality holds, the hidden layer is additive.

Axiom 3.10 (Identity tightness)**.** *The identity fiber is nonempty and zero-cost:*

$$\Pi^{-1}(e) \neq \emptyset, \qquad \forall \xi \in \Pi^{-1}(e),\ W(\xi) = 0.$$

3.3 Envelope lemmas

Lemma 3.11 (Envelope soundness)**.** *If $\xi \in \mathcal{G}$ and $\Pi(\xi)$ is admissible, then*

$$W(\xi) \leq \delta(\Pi(\xi)).$$

Proof. Let $R = \Pi(\xi)$. Then $\xi \in \mathcal{F}_R$ by definition. Since $\delta(R)$ is the supremum of W over $\mathcal{F}_R$, it is an upper bound for $W(\xi)$. □

Lemma 3.12 (Envelope minimality)**.** *Let Δ be any function on admissible observable traces satisfying*

$$W(\xi) \leq \Delta(\Pi(\xi))$$

for every hidden realization ξ whose projection is admissible. Then

$$\delta(R) \leq \Delta(R)$$

for every admissible trace R.

Proof. Fix R. For every $\xi \in \mathcal{F}_R$, one has $\Pi(\xi) = R$, hence

$$W(\xi) \leq \Delta(R).$$

Thus $\Delta(R)$ is an upper bound of $\{W(\xi) : \xi \in \mathcal{F}_R\}$. Since $\delta(R)$ is the least upper bound, $\delta(R) \leq \Delta(R)$. □

Lemma 3.13 (Identity envelope)**.**

$$\delta(e) = 0.$$

Proof. By identity tightness, the identity fiber is nonempty and every element of it has hidden cost zero. Therefore the set of costs over the identity fiber is $\{0\}$, whose supremum is zero. □

4 Projection-Induced Oplax Composition

Theorem 4.1 (Projection-induced oplax law)**.** *Let R_1, R_2 be admissible observable traces and assume $R_2 \odot R_1$ is admissible. Then*

$$\boxed{\delta(R_2 \odot R_1) \leq \delta(R_1) + \delta(R_2).}$$

Proof. Let $\xi \in \mathcal{F}_{R_2 \odot R_1}$. Then

$$\Pi(\xi) = R_2 \odot R_1.$$

By fiber decomposition, there exist $\xi_1, \xi_2 \in \mathcal{G}$ such that

$$\xi = \xi_2 \cdot \xi_1, \qquad \Pi(\xi_1) = R_1, \qquad \Pi(\xi_2) = R_2.$$

Thus $\xi_1 \in \mathcal{F}_{R_1}$ and $\xi_2 \in \mathcal{F}_{R_2}$. By hidden cost subadditivity,

$$W(\xi) = W(\xi_2 \cdot \xi_1) \leq W(\xi_1) + W(\xi_2).$$

Since $\xi_i \in \mathcal{F}_{R_i}$,

$$W(\xi_1) \leq \delta(R_1), \qquad W(\xi_2) \leq \delta(R_2).$$

Therefore

$$W(\xi) \leq \delta(R_1) + \delta(R_2).$$

Because ξ was arbitrary in the composite fiber, the right-hand side is an upper bound for all hidden costs over $\mathcal{F}_{R_2 \odot R_1}$. Taking the supremum gives

$$\delta(R_2 \odot R_1) \leq \delta(R_1) + \delta(R_2).$$

□

Remark 4.2 (Oplax, not additive)**.** *The theorem gives an inequality. Equality is allowed but not required. Projection may merge hidden realizations or destroy information, so the observable cost law is subadditive rather than strictly additive.*

5 V2 Certified Morphisms

Definition 5.1 (Coh boundary)**.** *A Coh boundary object X consists of a carrier Σ_X and a valuation*

$$V_X : \Sigma_X \to \mathbb{R}_{\geq 0}.$$

Definition 5.2 (Certified morphism)**.** *A certified morphism from $x \in \Sigma_X$ to $y \in \Sigma_Y$ is a tuple*

$$f = (R_f, s_f, d_f, x, y)$$

such that

$$R_f : x \rightsquigarrow y,$$
$$d_f \geq \delta(R_f),$$

and

$$V_Y(y) + s_f \leq V_X(x) + d_f.$$

Here s_f is declared spend and d_f is declared defect.

Definition 5.3 (Certified composition)**.** *For*

$$f = (R_f, s_f, d_f, x, y), \qquad g = (R_g, s_g, d_g, y, z),$$

define

$$g \circ f := (R_g \odot R_f,\ s_f + s_g,\ d_f + d_g,\ x, z)$$

whenever $R_g \odot R_f$ is defined, admissible, and typed $x \rightsquigarrow z$.

Theorem 5.4 (Closure of certified composition)**.** *If f and g are certified and their composite trace is defined, admissible, and correctly typed, then $g \circ f$ is certified.*

Proof. The endpoint and trace typing conditions hold by hypothesis. The defect condition follows from the oplax law:

$$\delta(R_g \odot R_f) \leq \delta(R_f) + \delta(R_g).$$

Since f and g are certified,

$$d_f \geq \delta(R_f), \qquad d_g \geq \delta(R_g).$$

Thus

$$d_f + d_g \geq \delta(R_f) + \delta(R_g) \geq \delta(R_g \odot R_f).$$

For the valuation inequality, certification gives

$$V_Y(y) + s_f \leq V_X(x) + d_f,$$

and

$$V_Z(z) + s_g \leq V_Y(y) + d_g.$$

Adding and canceling the finite term $V_Y(y)$ yields

$$V_Z(z) + (s_f + s_g) \leq V_X(x) + (d_f + d_g).$$

Therefore the composite is certified. □

Theorem 5.5 (V2 certified category)**.** *Certified boundary states and certified morphisms form a category* $\mathbf{Coh}_\delta$*.*

Proof. Closure follows from the closure theorem above. Identities are the zero-spend, zero-defect identity traces $\mathrm{id}_x = (e, 0, 0, x, x)$, certified by $\delta(e) = 0$ and $V_X(x) + 0 \leq V_X(x) + 0$. Associativity follows from associativity of observable trace composition and associativity of addition for spend and defect. Unit laws follow from the observable identity trace and zero spend/defect. □

6 V3: Induced Lawvere Geometry

Definition 6.1 (Admissible trace domain)**.** *For boundary states x, y, define*

$$\mathcal{T}(x, y) := \{R \in \mathcal{V}_{\mathrm{adm}} : R : x \rightsquigarrow y\}.$$

Equivalently, $\mathcal{T}(x, y)$ is the projection of the certified hom-class onto its trace component.

Definition 6.2 (Coh distance)**.** *Define*

$$\boxed{d_{\mathrm{Coh}}(x,y) := \inf_{R\in\mathcal{T}(x,y)} \delta(R)}$$

when $\mathcal{T}(x,y)$ *is nonempty, and set*

$$d_{\mathrm{Coh}}(x,y) = +\infty$$

when $\mathcal{T}(x,y) = \emptyset$*.*

Lemma 6.3 (Nonnegativity)**.** *For all states* x, y*,*

$$d_{\mathrm{Coh}}(x,y) \geq 0.$$

Proof. Each $\delta(R)$ is a supremum of nonnegative hidden costs, hence nonnegative. The infimum of nonnegative values is nonnegative, and the empty case is defined as $+\infty$. □

Lemma 6.4 (Identity distance)**.** *For every state* x*,*

$$d_{\mathrm{Coh}}(x,x) = 0.$$

Proof. The identity trace $e : x \rightsquigarrow x$ is admissible and satisfies $\delta(e) = 0$. Therefore

$$d_{\mathrm{Coh}}(x,x) \leq 0.$$

By nonnegativity, $d_{\mathrm{Coh}}(x,x) \geq 0$. Hence $d_{\mathrm{Coh}}(x,x) = 0$. □

Lemma 6.5 (Envelope dominates induced distance)**.** *If* $R \in \mathcal{T}(x,y)$*, then*

$$d_{\mathrm{Coh}}(x,y) \leq \delta(R).$$

Proof. The value $d_{\mathrm{Coh}}(x,y)$ is the infimum over all $\delta(S)$ with $S \in \mathcal{T}(x,y)$. Since R is one such trace, the infimum is no larger than $\delta(R)$. □

Theorem 6.6 (Directed triangle inequality)**.** *For all states* x, y, z*,*

$$\boxed{d_{\mathrm{Coh}}(x,z) \leq d_{\mathrm{Coh}}(x,y) + d_{\mathrm{Coh}}(y,z).}$$

Proof. If either $d_{\mathrm{Coh}}(x,y)$ or $d_{\mathrm{Coh}}(y,z)$ is $+\infty$, the right-hand side is $+\infty$ and the claim is immediate. Otherwise fix $\varepsilon > 0$. By the definition of infimum, choose traces

$$R_1 \in \mathcal{T}(x,y), \qquad R_2 \in \mathcal{T}(y,z)$$

such that

$$\delta(R_1) \leq d_{\mathrm{Coh}}(x,y) + \varepsilon/2,$$

and

$$\delta(R_2) \leq d_{\mathrm{Coh}}(y,z) + \varepsilon/2.$$

By trace closure, $R_2 \odot R_1 \in \mathcal{T}(x,z)$. Hence

$$d_{\mathrm{Coh}}(x,z) \leq \delta(R_2 \odot R_1).$$

By the oplax law,

$$\delta(R_2 \odot R_1) \leq \delta(R_1) + \delta(R_2).$$

Therefore

$$d_{\mathrm{Coh}}(x,z) \leq d_{\mathrm{Coh}}(x,y) + d_{\mathrm{Coh}}(y,z) + \varepsilon.$$

Letting $\varepsilon \to 0$ yields the triangle inequality. □

Theorem 6.7 (Lawvere enrichment)**.** *The state-indexed Coh structure carries a directed extended Lawvere quasi-metric over the monoidal poset*

$$([0,\infty], \geq, +, 0).$$

Proof. The identity law is $d_{\mathrm{Coh}}(x,x) = 0$. The enriched composition law in the reversed order $\geq$ is

$$d_{\mathrm{Coh}}(x,y) + d_{\mathrm{Coh}}(y,z) \geq d_{\mathrm{Coh}}(x,z),$$

which is exactly the directed triangle inequality. Symmetry is not assumed and generally fails. □

7 The Coh Bridge Theorem

Theorem 7.1 (Coh Bridge Theorem)**.** *Let a Coh system satisfy:*

1. *V1 verifier identity and verifier closure under trace concatenation;*
2. *additive spend and defect bookkeeping;*
3. *finite cancellable valuations;*
4. *a V2 projection layer $\Pi : \mathcal{G} \to \mathcal{V}$;*
5. *nonempty bounded hidden fibers for admitted traces;*
6. *projection compatibility;*
7. *fiber decomposition;*
8. *hidden cost subadditivity;*
9. *identity tightness $\delta(e) = 0$.*

Then:

1. *V1-admissible traces form a category;*
2. *the V2 envelope defect satisfies the oplax law*
$$\delta(R_2 \odot R_1) \leq \delta(R_1) + \delta(R_2);$$
3. *V2 certified morphisms form a category $\mathbf{Coh}_\delta$;*
4. *the V3 function*
$$d_{\mathrm{Coh}}(x,y) = \inf_{R \in \mathcal{T}(x,y)} \delta(R)$$
defines a directed extended Lawvere quasi-metric.

Proof. Part (1) is the V1 category theorem. Part (2) is the projection-induced oplax theorem. Part (3) follows because the composite declared defect $d_f + d_g$ dominates the envelope of the composite trace by the oplax law, and because the endpoint resource inequalities compose by cancellation of the intermediate valuation. Part (4) follows from identity tightness and the directed triangle inequality proved above. Hence verifier admissibility induces a projection-certified category, and the projection-certified category induces Lawvere-style directed geometry. □

8 Interpretation

The bridge theorem shows that the Coh layers are not independent manuscripts. They form a structural pipeline:

$$\boxed{\text{local verifier law} \Longrightarrow \text{certified category} \Longrightarrow \text{projection envelope} \Longrightarrow \text{directed geometry.}}$$

V1 answers: which traces are allowed? V2 answers: what hidden cost must an observable trace conservatively dominate? V3 answers: what is the least admissible projection cost from one state to another?

The key identity is:

$$\boxed{\text{V3 triangle inequality} = \text{V2 oplax defect law under infimum.}}$$

Thus Coh geometry is not added externally; it is induced by admissibility and projection.

9 Failure Modes

Remark 9.1 (Verifier closure failure)**.** *If verifier-accepted traces are not closed under concatenation, V1 does not form a category. The structure may be a graph or partial category, but the full bridge to V3 fails.*

Remark 9.2 (Fiber decomposition failure)**.** *If a hidden realization of a composite observable trace cannot be decomposed into realizations of the components, the oplax theorem fails. For non-Markovian or entangled systems, this must be replaced by a weak glue-defect law.*

Remark 9.3 (Unbounded fibers)**.** *If* $\sup_{\xi \in \mathcal{F}_R} W(\xi) = +\infty$*, then* $\delta(R) = +\infty$*. The metric remains extended-valued, but finite certification requires a finite computable envelope.*

Remark 9.4 (Degenerate defect)**.** *If declared defect is arbitrarily loose, the category may remain formally valid but lose selectivity. The induced geometry may collapse because too many transitions become cheaply admissible.*

Remark 9.5 (No identity tightness)**.** *If* $\delta(e) \neq 0$*, then* $d_{\mathrm{Coh}}(x, x) = 0$ *is not guaranteed. Identity tightness is therefore part of the bridge specification, not optional polish.*

10 Worked Separation Example

Let $X = \{A, B, C\}$ and let

$$A \xrightarrow{\sigma_1} B \xrightarrow{\sigma_2} C$$

be a two-step observable trace. Suppose

$$s(\sigma_1) = 1, \qquad s(\sigma_2) = 1.$$

Then the syntactic spend of the composite is

$$s_{\mathrm{syn}}(\sigma_2 * \sigma_1) = 2.$$

Suppose the hidden fiber of the composite contains a single hidden representative Θ^* with

$$W(\Theta^*) = 1.$$

Then

$$\delta(\sigma_2 \odot \sigma_1) = 1 < 2 = s_{\mathrm{syn}}(\sigma_2 * \sigma_1).$$

This illustrates that semantic envelope cost and syntactic bookkeeping cost are distinct channels.

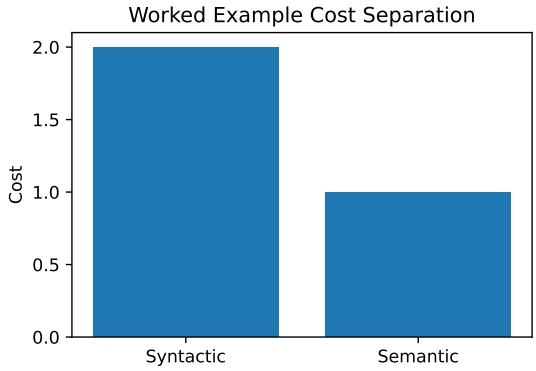


Figure 4: Worked separation: syntactic bookkeeping can exceed semantic envelope cost after projection through hidden realizations.

11 Lean Alignment Sketch

A direct Lean implementation should separate data from proof obligations:

1. states and endpoint-typed traces;
2. verifier acceptance as a Boolean with a correctness theorem or as a proposition;
3. admissible morphisms as dependent records carrying verifier and inequality proofs;
4. hidden realizations and observable traces;
5. projection compatibility and fiber decomposition as theorem obligations;
6. envelope defect as a supremum in the analytic model or as a conservative computable upper bound in executable models;
7. distance as an infimum over endpoint-indexed admissible traces.

For V2 strictness, fiber decomposition should be implemented using segmentable witness records. For V3/V4 weak dynamics, a glue defect ϵ_{glue} should be added when exact decomposition fails.

12 Conclusion

The Coh Bridge Theorem closes the conceptual gap between the operational, categorical, and geometric layers. V1 supplies certified admissible traces. V2 supplies projection-induced envelope defect and the oplax law. V3 takes the infimum of this envelope over admissible traces and obtains directed Lawvere geometry.

The safe structural claim is:

$$\boxed{\text{Verifier admissibility plus projection-aware cost induces categorical geometry.}}$$

This is the bridge result.

References

[1] F. W. Lawvere, *Metric spaces, generalized logic, and closed categories*, Rendiconti del Seminario Matematico e Fisico di Milano 43 (1973), 135–166.

[2] G. M. Kelly, *Basic Concepts of Enriched Category Theory*, Cambridge University Press, 1982.

[3] J. Bénabou, *Introduction to bicategories*, Reports of the Midwest Category Seminar, Springer, 1967, 1–77.

[4] T. Leinster, *Basic Category Theory*, Cambridge University Press, 2014.